TOPIC 2 Systems of Linear Equations and Inequalities

Name _____ Date _____

Reminders

Intercepts: set y = 0 to find the x-intercept; set x = 0 to find the y-intercept.

Solution of a system: the point where the lines cross. It makes both equations true.

How many? Different slopes: one. Same slope, different y-intercept: none. Same line: infinite.

1. How many solutions does the system $y = 2x + 5$ and $y = 2x - 1$ have?

Tip: Compare the slopes, then the y-intercepts.

A. One solution

B. Infinitely many

C. No solution

2. Which system has infinitely many solutions?

Tip: Write both in slope-intercept form.

F. $y = 3x + 2$ and $2y = 6x + 4$

G. $y = 3x + 2$ and $y = 3x - 2$

H. $y = 3x + 2$ and $y = -3x + 2$

3. Maya has $20 and saves $5 a week. Leo also has $20 and saves $5 a week. Write a system and tell how many solutions it has.

> **Tip:** Start amount is b, weekly amount is m.

Maya: y = ____x + ____

Leo: y = ____x + ____

Same line, so _____ solutions